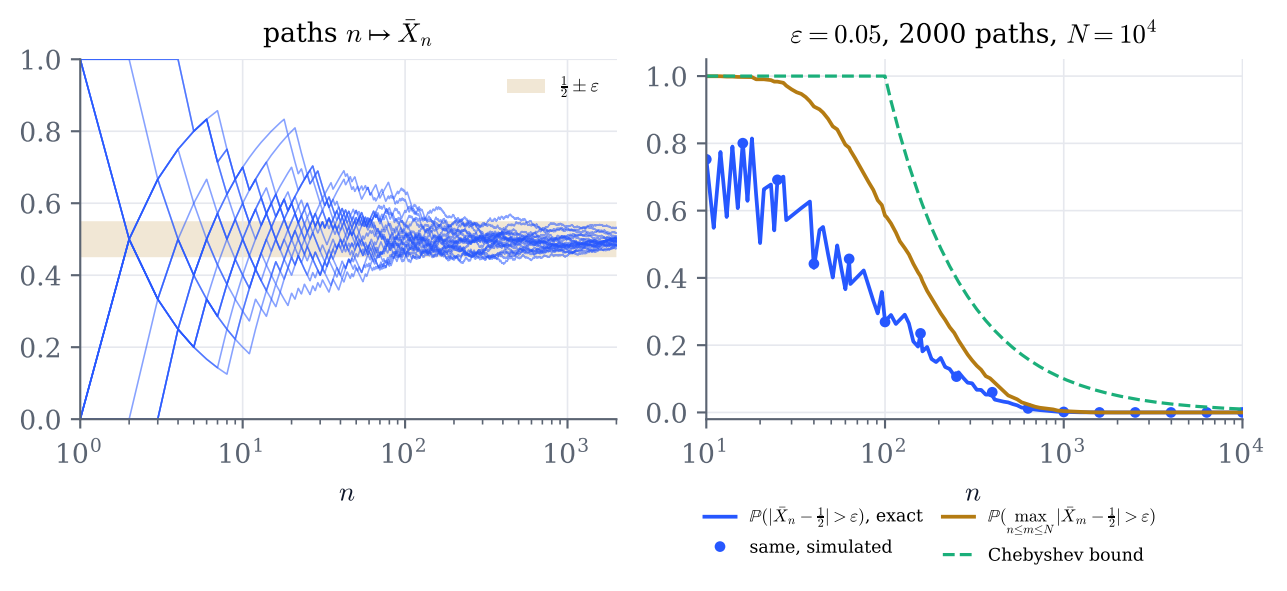

paths $n \mapsto \bar{X}_n$
$\frac{1}{2} \pm \varepsilon$
1.0
0.8
0.6
0.4
0.2
0.0
10^0
10^1
10^2
10^3
n
$\varepsilon = 0.05$, 2000 paths, $N = 10^4$
1.0
0.8
0.6
0.4
0.2
0.0
10^1
10^2
10^3
10^4
n
$\mathbb{P}(|\bar{X}_n - \frac{1}{2}| > \varepsilon)$, exact
same, simulated
$\mathbb{P}(\max_{n \le m \le N} |\bar{X}_m - \frac{1}{2}| > \varepsilon)$
Chebyshev bound